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3. Chlorophyll 'd' differs from Chlorophyll 'a' in having O-CHO group instead of CH-CH_2 group at 2nd Carbon in the 1st Pyrrol ring.
4. Pheophytin resembles Chlorophyll 'a' except that it lacks Mg atom. Instead it has two H atoms.
5. Phycobilins have open tetra pyrrols and they have neither Mg nor phytol chain.

13.3.2 Carotenoids

Carotenoids are yellow to orange pigments, mostly tetraterpens and these pigments absorb light strongly in the blue to violet region of visible spectrum. These pigments protect chlorophyll from photo-oxidative damage. Hence, they are called as **shield pigments**. These pigments absorb light and transfer these to chlorophyll. Almost all carotenoid pigments have 40 carbon atoms. Ripening of fruits, floral colours and leaf colour change during autumn is due to Carotenoids (Carotene and Xanthophyll) (Figure 13.3).

i. Carotenes:

Orange, Red, Yellow and Brownish pigments, hydrocarbons (Lipids) and most of them are tetraterpenes($C_{40}H_{56}$). Carotene

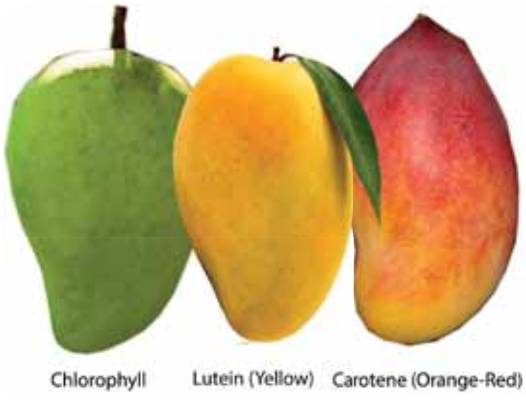


Figure 13.3: Changes in Fruit colour due to difference in pigmentation

Separation of Chloroplast pigments by paper Chromatography method

Step 1. Extract chlorophyll pigment from the leaves using 80% Acetone.

Step 2. Allow to concentrate by evaporation.

Step 3. Apply few drops on one end above 2 cm from the edge of a chromatographic paper.

Step 4. A solvent with mixture of Petroleum ether and acetone in the ratio of 9:1 is prepared and poured into development chamber.

Step 5. Place the strip above the solvent by placing one end of the strip touching the solvent.

Observation

After one hour observe the chromatographic paper. You can find the pigments being separated into four distinct spots (Figure 13. 4).

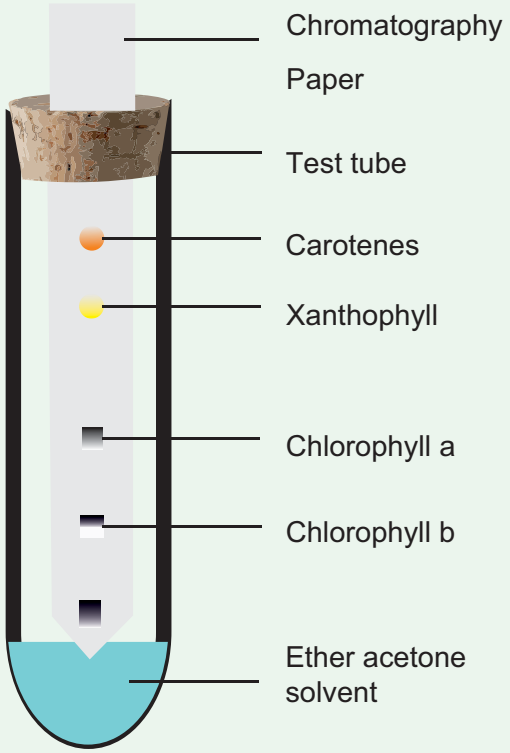


Figure 13.4: Paper Chromatography